EXP NO1: 6

HAMMING CODE

AIM

Write a program to implement error detection and correction using hamming code concept. Make a list sum to input data stream and verify error correction features

Create sender program with below features:

1. Input to sender file should be a list of any length. Program should convert the text to binary.

2. Apply hamming code concept one binary data and add redundant bits to it.

3. save this output in an array, display the position of the error.

4. Else somewhere remove the redundant bits and convert the binary data to ascii and display output.

Create a receiver program with below features:

1. Receive program should read the input from channel of it.

2. Apply hamming code on the binary to check for errors

3. If there is an error, display the position of the error

4. Else somewhere remove the redundant bits and convert the binary data to ascii and display output.

CODE

Sender Program

```
s= input ("Enter text:")
b= ''.join(format(ord(c),'08b') for c in s)
m= len(b)
r= 0
while 2**r < m+r+1:
    r+=1
a= [0] * (m+r+1)
j=0
for i in range (1, m+r+1):
    if i & (i-1):
        a[i] = b[j]
        j+=1
for p in range (r):
    k= 2**p
    a[k] = str (sum (int(a[i]) for i in range (1, len(a)) if i & k))
code = ''.join (a[1:])
print ("Hamming code :", code)
```

Receiver Program:

```
c= list (open ('
n= len (c)
error= 0
p= 1
while p <
    parity
    if 1
```

Sample

Sample

RESULT:

Thus

output

Receiver Program:

```
c= list (open ("channel.txt").read())
n= len (c)
error=0
p=1
while p<=n
    parity = sum (int (c[i-1]) for i in range (1,n+1)
    if i&p)%2
        if parity:
            error += p
            p*=2
    if error
        print ("Error at position: ", error)
        c[error-1] = '1' if c[error-1]=='0' else '0'
        print ("Error corrected")
    else:
        print ("No error")
    data = ''.join (c[i-1] for i in range (1,n+1) if i&(i-1))
    out = ''.join (chr (int (data[i:i+8],2)) for i in range (0,
                    len (data),8))
    print ("output - ",out)
```

Sample input:

HELLO

Sample output:

Hamming code: 1001100010...
No errors
output: HELLO

RESULT:

Thus the code is executed successfully and output is verified.